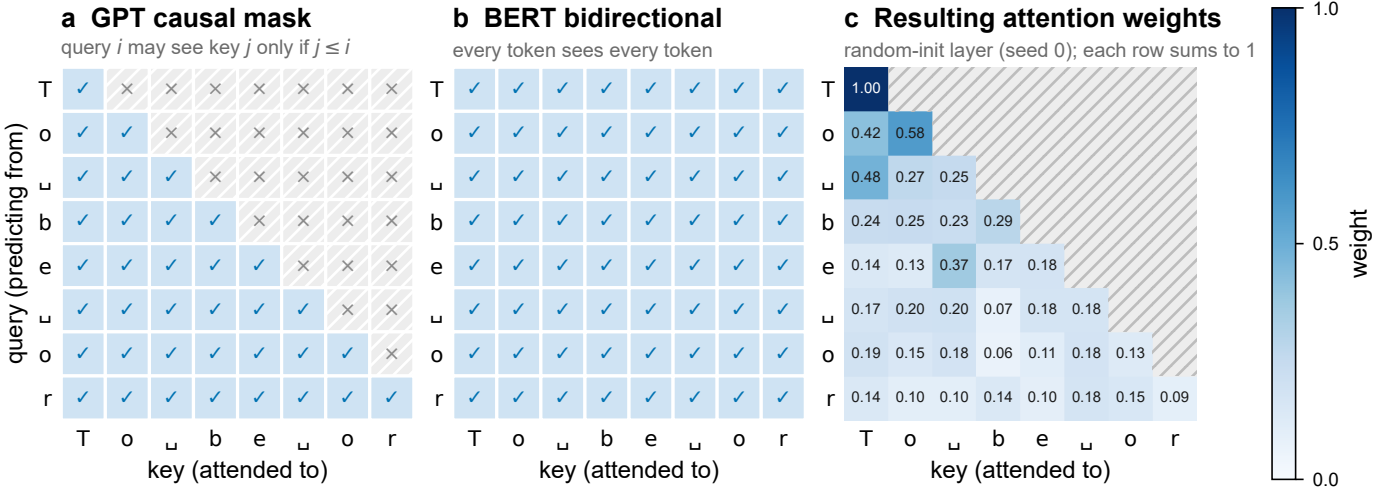

a GPT causal mask
query i may see key j only if j ≤ i
b BERT bidirectional
every token sees every token
c Resulting attention weights
random-init layer (seed 0); each row sums to 1
query (predicting from)
key (attended to)
T o ␣ b e ␣ o r
1.00
0.42 0.58
0.48 0.27 0.25
0.24 0.25 0.23 0.29
0.14 0.13 0.37 0.17 0.18
0.17 0.20 0.20 0.07 0.18 0.18
0.19 0.15 0.18 0.06 0.11 0.18 0.13
0.14 0.10 0.10 0.14 0.10 0.18 0.15 0.09
weight
1.0
0.5
0.0